

CloudExify Cybersecurity Internship

Project 04

Vulnerability Assessment & Remediation

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❖ <u>Assessment Tool</u>	<u>Tenable Nessus Essentials</u>
❖ <u>Assessment Type</u>	<u>Authorized Vulnerability</u>
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➤ Introduction

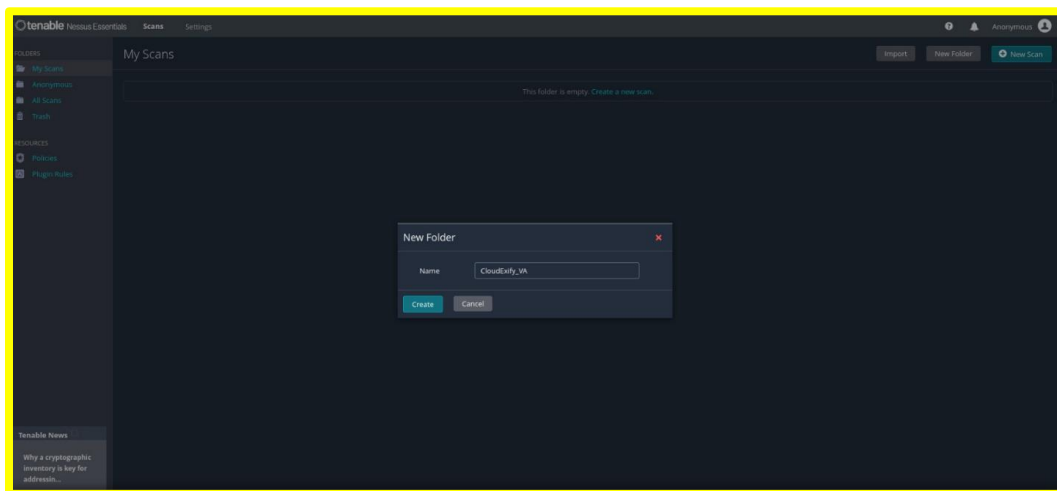
Vulnerability Assessment is an important part of cybersecurity that helps identify, analyze, and evaluate security weaknesses in a system or network. The purpose of this project is to perform a structured vulnerability assessment using **Tenable Nessus Essentials** and document the identified security findings in a professional manner.

This project follows a systematic assessment process, including scanning the target system, identifying vulnerabilities, reviewing detected services and software, analyzing the severity of findings, and preparing suitable remediation recommendations. Nessus was used as the primary assessment tool to identify potential security issues and provide detailed information about the target host.

The assessment was conducted as an authorized vulnerability assessment under the CloudExify Cybersecurity Internship. The results were documented with screenshots and technical details to provide evidence of the assessment and support further security improvement. The project demonstrates practical skills in vulnerability scanning, security analysis, risk assessment, documentation, and remediation planning.

➤ Creating a Dedicated Folder for the Vulnerability Assessment

Screenshot



Purpose:

To create a dedicated folder for organizing the vulnerability assessment scan.

Objective:

To maintain the Nessus scan in a structured and easily accessible location.

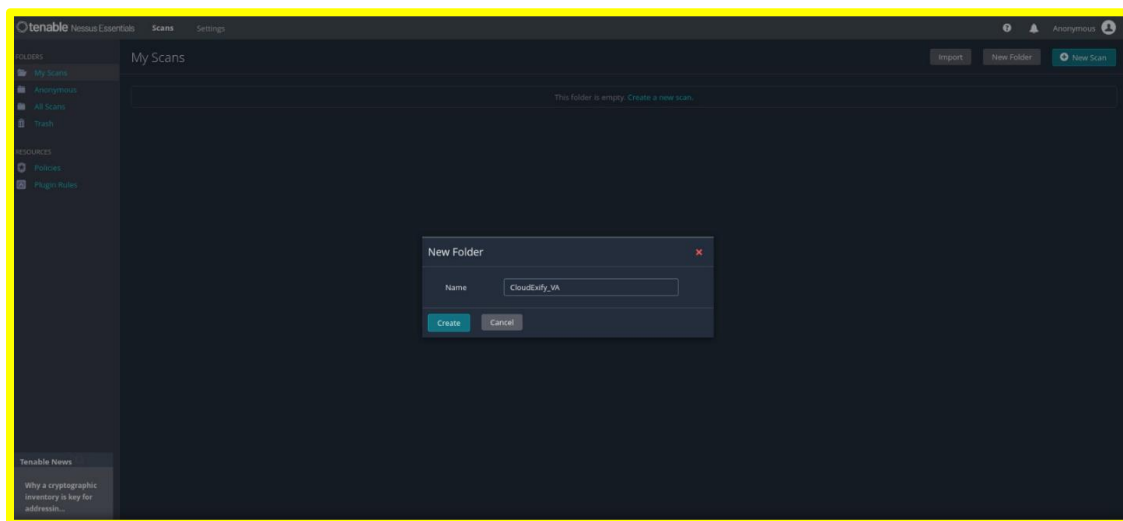
Description:

The Nessus Essentials interface displays the **New Folder** dialog box with the folder name **CloudExify_VA** entered. The folder is intended to store the vulnerability assessment scan.

Result:

The **CloudExify_VA** folder was created successfully for organizing the assessment.

➤ **Creating the CloudExify Vulnerability Assessment Scan**

Screenshot**Purpose:**

To create a new Nessus vulnerability assessment scan within the dedicated project folder.

Objective:

To prepare the Nessus scan before initiating the vulnerability assessment.

Description:

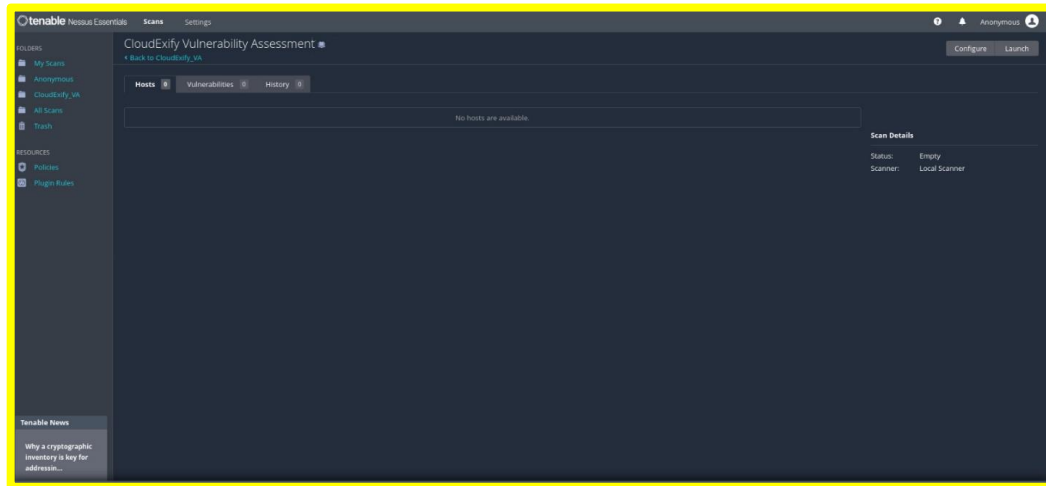
A Nessus scan named **CloudExify Vulnerability Assessment** was created. The scan interface initially shows no hosts or vulnerabilities because the scan had not yet been executed.

Result:

The vulnerability assessment scan was successfully created and prepared for configuration and execution.

➤ Vulnerability Assessment Scan in Progress

Screenshot



Purpose:

To initiate the Nessus vulnerability assessment against the configured target.

Objective:

To identify potential security weaknesses and vulnerabilities in the target system.

Description:

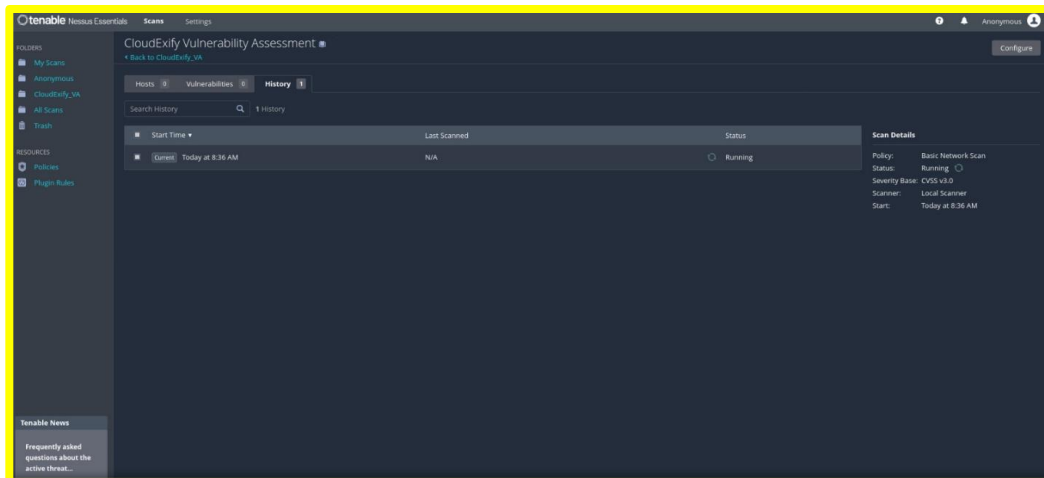
The Nessus scan is shown running under the **History** tab. The scan uses the **Basic Network Scan** policy and the **Local Scanner**. The status is displayed as **Running**, indicating that Nessus is actively performing the assessment.

Result:

The vulnerability assessment started successfully and was actively scanning the target.

➤ Vulnerabilities Detected During the Scan

Screenshot



Purpose:

To monitor the vulnerabilities identified while the Nessus assessment is running.

Objective:

To observe the security findings discovered during the vulnerability scanning process.

Description:

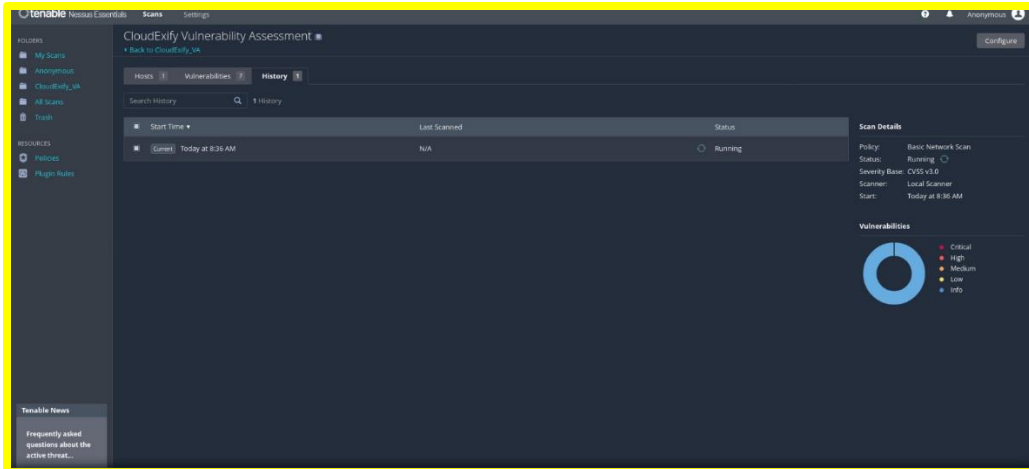
The Nessus interface shows the scan in progress with **1 host** and **7 vulnerabilities** detected at this stage. The vulnerability summary is displayed on the right side of the interface.

Result:

Nessus successfully identified initial vulnerabilities during the ongoing assessment.

➤ Completed Vulnerability Assessment

Screenshot



Purpose:

To verify the completion of the Nessus vulnerability assessment.

Objective:

To obtain the final vulnerability assessment results for analysis and reporting.

Description:

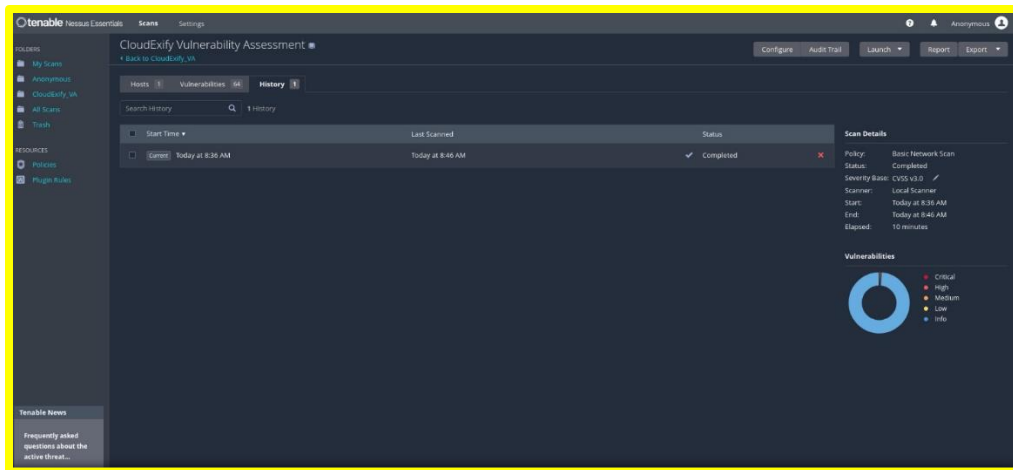
The Nessus scan is displayed with a **Completed** status. The assessment scanned **1 host** and identified **64 vulnerabilities**. The scan used the **Basic Network Scan** policy and completed in approximately **10 minutes**.

Result:

The vulnerability assessment was completed successfully, producing **64 vulnerability findings** for further analysis and remediation.

➤ Final Nessus Scan History and Vulnerability Summary

Screenshot



Purpose:

To document the final scan status and overall vulnerability findings.

Objective:

To preserve evidence of the completed vulnerability assessment for the project report.

Description:

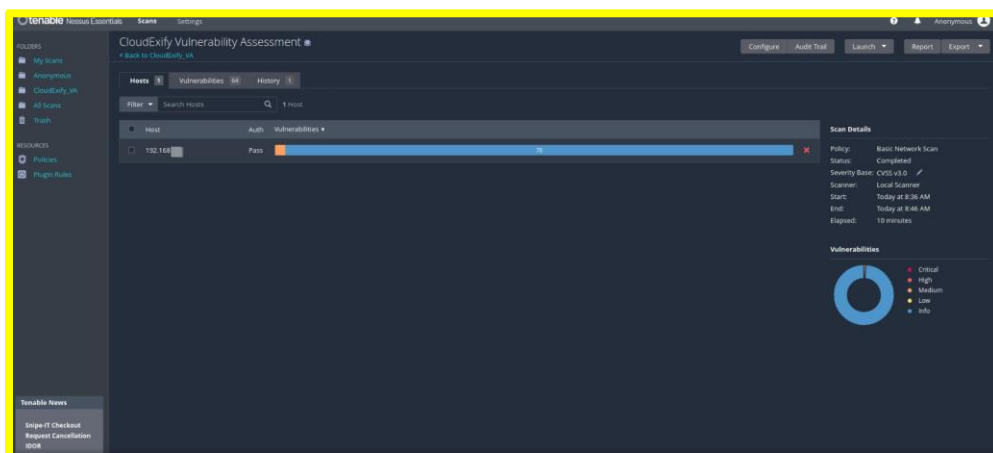
The Nessus **History** section shows the completed assessment, including the start and end times. The interface reports **64 vulnerabilities** and confirms that the scan status is **Completed**.

Result:

The final Nessus scan results were successfully recorded and are ready for vulnerability analysis and reporting.

➤ Completed Nessus Vulnerability Assessment

Screenshot



Objective:

To identify and assess security issues present on the scanned system.

Description:

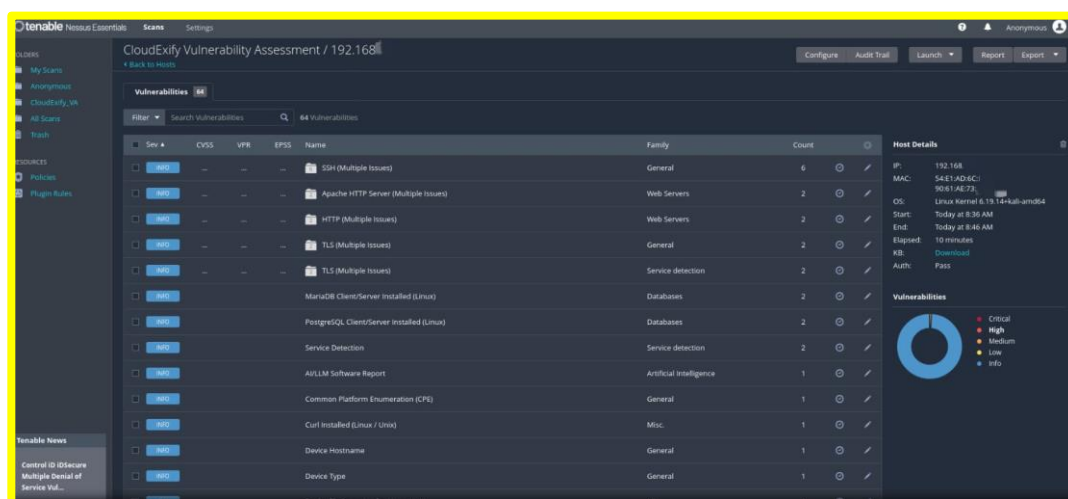
The screenshot shows the completed **CloudExify Vulnerability Assessment** in Nessus Essentials. The scan targeted a host and completed successfully using the **Basic Network Scan** policy. The results include the host status, authentication status, scan duration, and vulnerability summary.

Result:

The vulnerability assessment was completed successfully, and Nessus identified security-related findings on the target host.

➤ Vulnerability List of the Target Host

Screenshot

**Purpose:**

To examine the vulnerabilities detected on the target system.

Objective:

To review the security findings categorized by severity and vulnerability family.

Description:

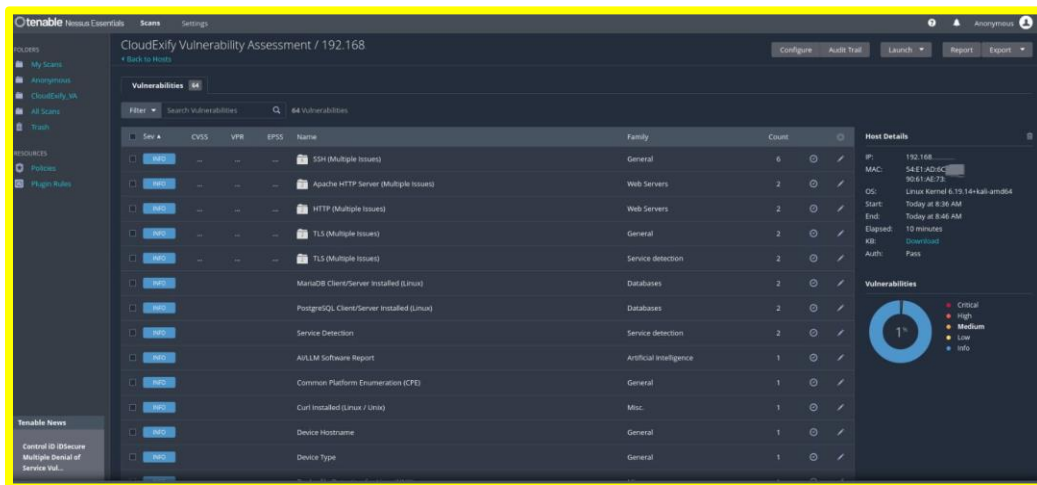
The screenshot displays the vulnerability list for the scanned host. Nessus reports **64 vulnerabilities/findings**, including entries related to SSH, Apache HTTP Server, HTTP, TLS, MariaDB, PostgreSQL, and service detection.

Result:

Multiple security findings were detected and successfully categorized by Nessus for further analysis.

➤ **Detected Services and Software**

Screenshot

**Purpose:**

To identify the services and software components detected on the target host.

Objective:

To determine which network services and software installations are exposed or present on the system.

Description:

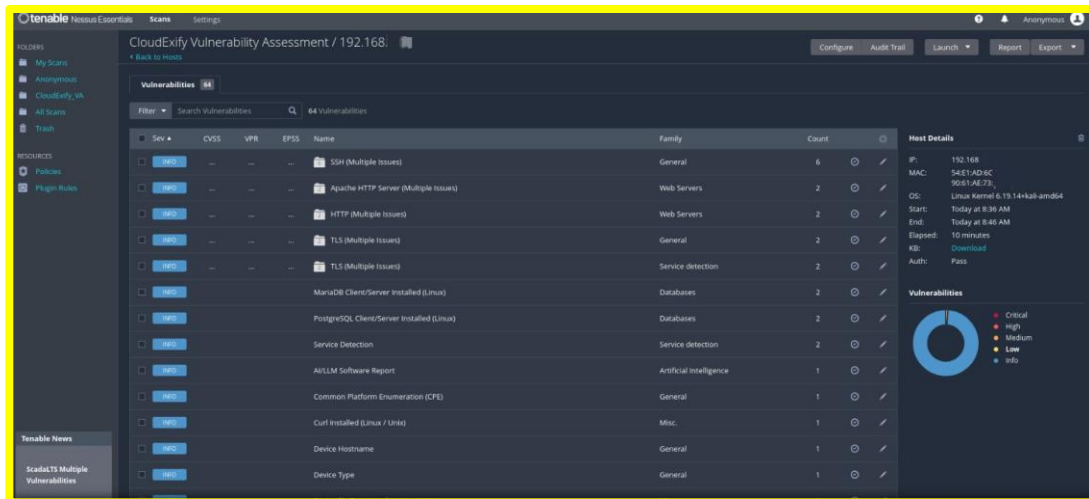
The Nessus results show findings related to **SSH, Apache HTTP Server, HTTP, TLS, MariaDB, PostgreSQL, and other system components**. These findings provide information about the host's running services and installed software.

Result:

Nessus successfully identified several services and software components running on the target system.

➤ **Host Security Assessment Details**

Screenshot



Purpose:

To review detailed information about the scanned host.

Objective:

To verify the target host information and authentication status recorded during the assessment.

Description:

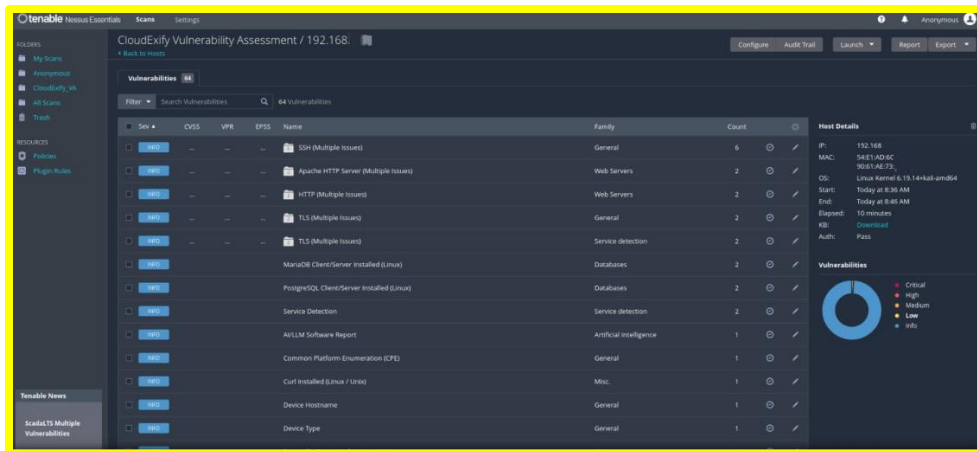
The Host Details panel displays information such as the target IP address, operating system, scan start and end times, elapsed time, and authentication status. The scan shows **Authentication: Pass**.

Result:

The target host was successfully assessed, and authenticated scanning was completed successfully.

➤ Vulnerability Severity Overview

Screenshot



Purpose:

To analyze the severity distribution of the findings.

Objective:

To understand the overall security condition of the scanned host based on Nessus severity categories.

Description:

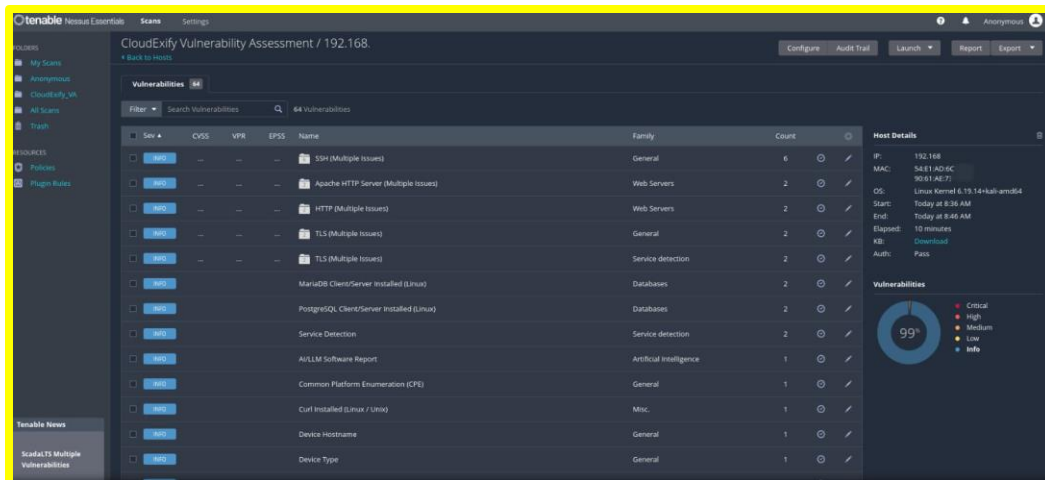
The vulnerability summary on the right side presents the findings according to severity levels: **Critical, High, Medium, Low, and Informational**. The displayed results are predominantly informational findings.

Result:

The scan provided a clear severity-based overview of the security findings detected on the host.

➤ Nessus Vulnerability Assessment Summary

Screenshot



Purpose:

To document the overall results of the vulnerability assessment.

Objective:

To provide a consolidated view of the vulnerabilities, host information, and security findings.

Description:

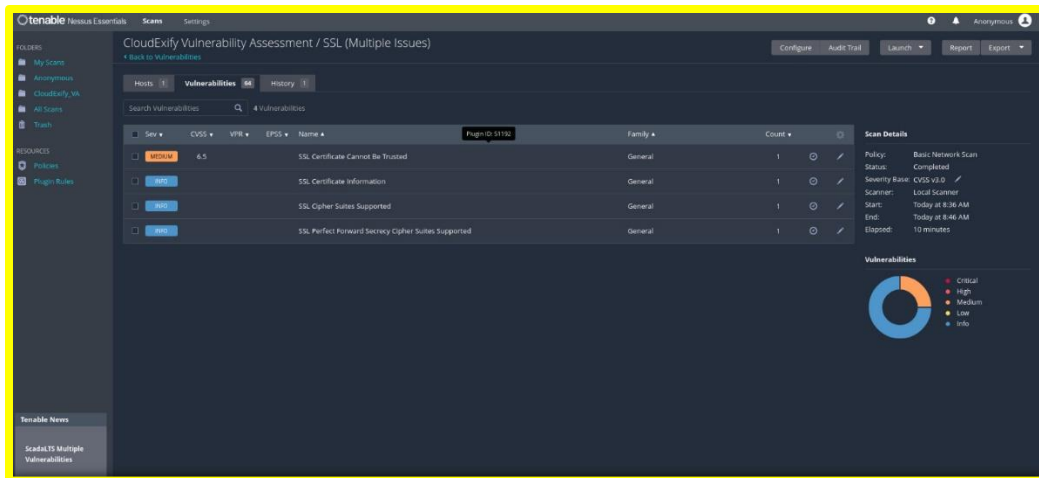
The screenshot represents the final Nessus assessment results, showing the vulnerability count, detected services, host details, authentication status, and severity distribution. This information can be used for further security analysis and reporting.

Result:

The Nessus assessment successfully generated a comprehensive set of findings for the target host, which can be analyzed to identify areas requiring security improvement.

➤ Vulnerability Overview - SSL Certificate Cannot Be Trusted

Screenshot



Purpose:

To document the vulnerability identified by Nessus in the target system.

Objective:

To review the severity, CVSS score, vulnerability name, and related SSL/TLS findings.

Description:

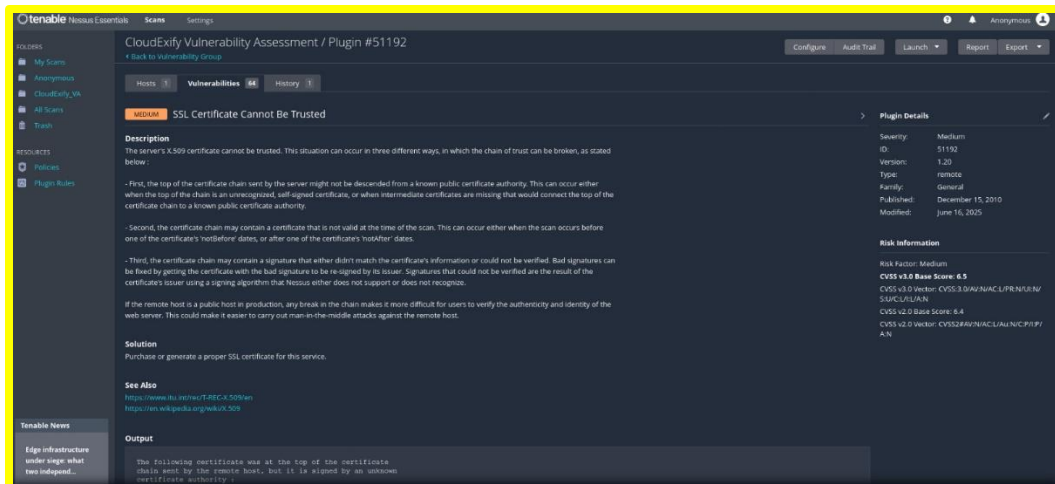
The Nessus vulnerability list displays four SSL-related findings. The main finding, **SSL Certificate Cannot Be Trusted**, is classified as **Medium severity** with a **CVSS score of 6.5**. The remaining findings are informational and relate to SSL certificate information, supported cipher suites, and Perfect Forward Secrecy cipher suites.

Result:

Nessus successfully identified the SSL certificate trust issue and categorized it as a **Medium-severity vulnerability**, while the other SSL-related findings were recorded as informational.

➤ Detailed Analysis of SSL Certificate Cannot Be Trusted

Screenshot



Purpose:

To examine the technical details of the identified SSL certificate vulnerability.

Objective:

To understand the vulnerability description, risk information, CVSS score, and recommended solution provided by Nessus.

Description:

The Nessus plugin details identify **SSL Certificate Cannot Be Trusted** as a **Medium-severity** vulnerability with a **CVSS v3.0 score of 6.5** and **Plugin ID 51192**. Nessus explains that the certificate trust chain may be broken due to an unrecognized certificate authority, an invalid certificate, or a certificate signature that cannot be verified.

Result:

The vulnerability was confirmed by Nessus with a **CVSS score of 6.5 (Medium)**. Nessus recommends purchasing or generating a proper SSL certificate for the affected service to resolve the certificate trust issue.

